

Computer System

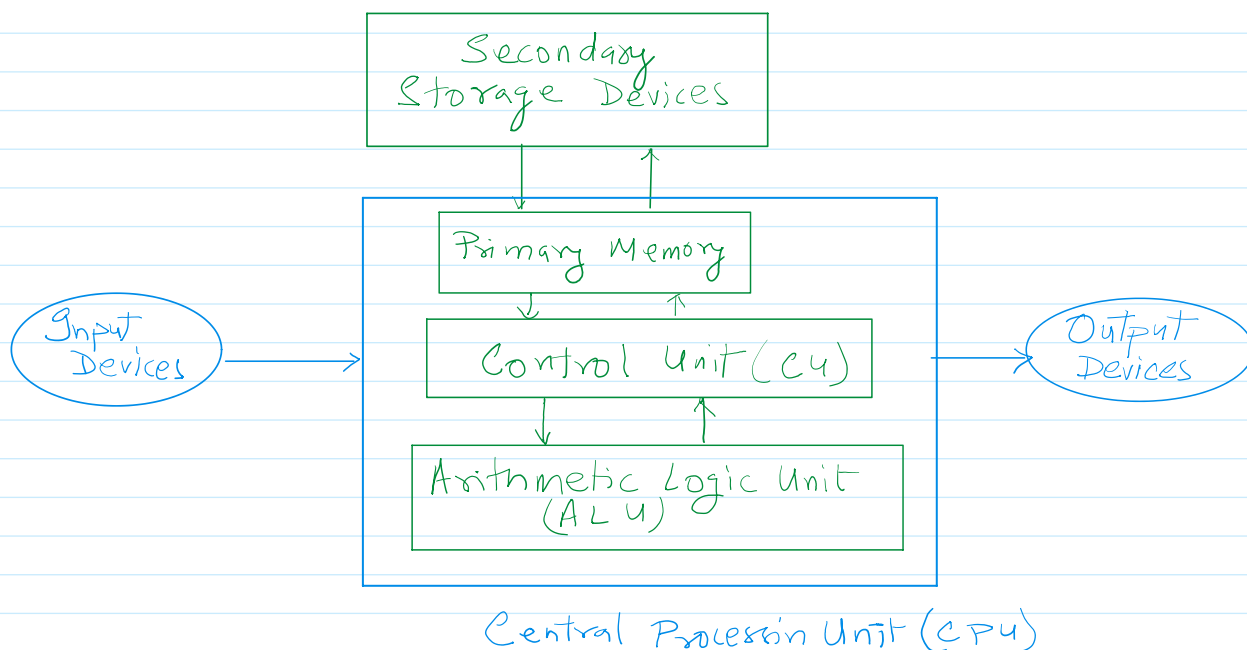
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A computer is an electronic device that can be programmed to accept data (input) process it, and generate a result (output).

What is computer system?

A computer along with additional hardware and software together is called a **computer system**.

A computer system primarily comprises a central processing unit (CPU) memory, Input/output devices and storage devices. All these components function together as a single unit to deliver the desired output. A computer system comes in various forms and sizes. It can vary from high end server to personal desktop.



Central Processing Unit(CPU)

It is the electronic circuitry of a computer that carries out the actual processing and usually referred as the brain of the computer. It is commonly called processor also. Physically, a CPU can be placed on one or more microchips called integrated circuits (IC). The ICs comprise semiconductor materials.

The CPU is given instruction and data through programs. The CPU then fetches the program and data from the memory and performs arithmetic and logic operations as per the given instructions and store the result back to memory.

While processing the CPU stores the data as well as instructions in its local memory called registers. Registers are part of the CPU chips and they are limited in size and number. Different registers are used for storing data instructions or intermediate results.

Other than registers, the CPU has two main components - Arithmetic logic unit (ALU) and control unit (CU). ALU performs all the arithmetic and logic operations that need to be done as per the instruction in a program. Control unit controls sequential instruction execution, interprets instructions, and guides data flow through the computer's memory ALU and input or output devices. CPU is also popularly known as microprocessors.



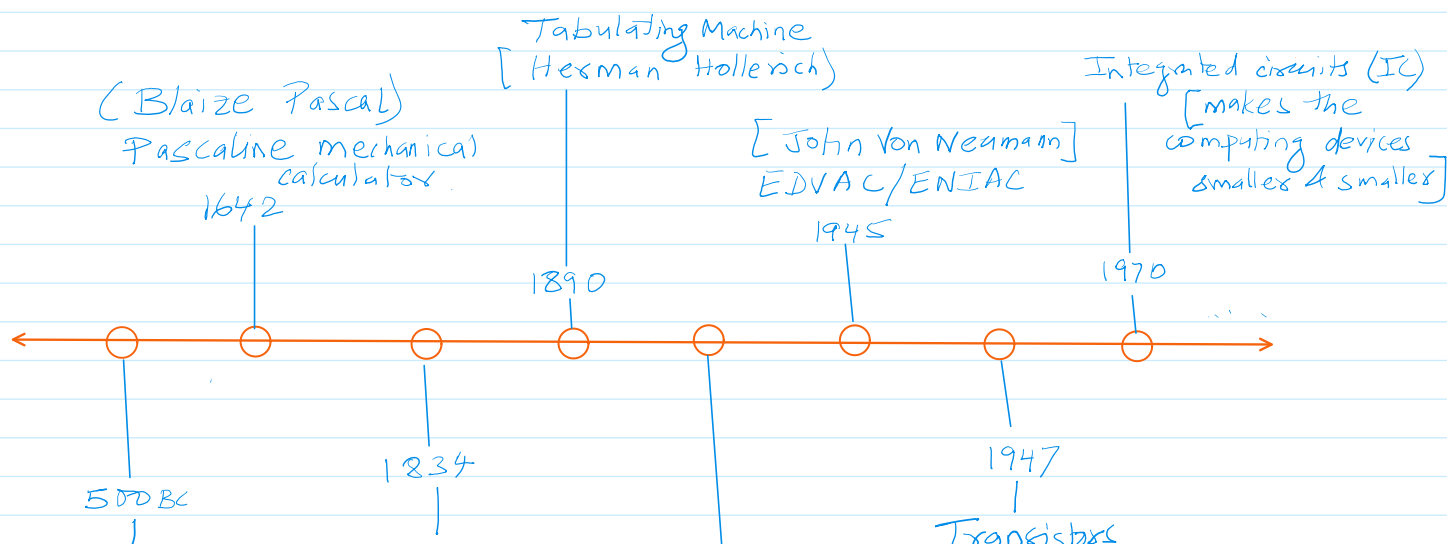
Input Devices: The devices through which control signals are sent to the computer are termed as input devices. These devices convert the input data into a digital form that is acceptable by the computer system. Some examples of input devices include keyboard, mouse, scanner, touch screen, etc.

Data entered through input devices is temporarily stored in main memory (also called RAM) of the computer system. For permanent storage and future use, the data as well as instructions are stored permanently in additional storage location called secondary memory.

Output devices: The device that receives data from a computer system for display, physical production, etc. Is called output devices. It converts digital information into human understandable form. For example, monitor, projector, headphones, speaker, printer, etc.

Evolution of computer

From the simple calculator to a modern day powerful data processor, computing devices have evolved in relatively short span of time. The evolution of computing devices is shown through a timeline. In the following figure:



1
500 BC
|
Abacus

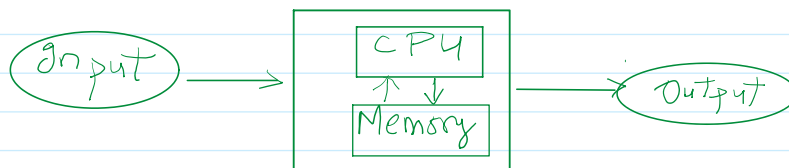
1834
|
Analytical Engine
(Charles Babbage)

1947
|
Transistors
[vacuum Tubes were replaced by transistors developed at Bell Labs, using semiconductor materials]

1937
|
Turing Machine

[concept was a general purpose programmable machine that was capable of solving any problem by executing the program stored on the punched cards.]

Timeline showing key inventions in computing technology.



Von Neumann architecture for the computer

ENIAC [Electronic Numerical Integrated And Computer]

Computer Memory

Computer system needs memory to store. The data and instructions for processing. Whenever we talk about memory of a computer system, we usually talk about the main or primary memory. The secondary memory (also called storage device) is used to store data, instructions and results permanently for future use.

Units of Memory

A computer system uses binary numbers to store and process data. The binary digits zero and one, which are the basic units of memory, are called bits. Further, these bits are grouped together to form words. A 4 - bit word is called a nibble. Example of nibbles are 1001, 1010, 0010 etc. A 2 nibble word that is eight - bit word is called a byte. For example. 01000110, 01111100 etc.

Like any other standard, unit bytes are grouped together to make bigger chunks of units of memory. The following table shows different measurement units for digital data stored in a storage device.

Measurement units for digital data

Unit	Description
KB(Kilobyte)	1 KB = 1024 Bytes
MB(Megabyte)	1 MB = 1024 KB

GB(Gigabyte)	1 GB = 1024 MB
TB(Terabyte)	1 TB = 1024 GB
PB(Petabyte)	1 PB = 1024 TB
EB(Exabyte)	1 EB = 1024 PB
ZB(Zetabyte)	1 ZB = 1024 PB
YB(Yottabyte)	1 YB = 1024 ZB

Type of Memory

Are human beings memorize many things over a lifetime and recall from memory to make a decision or some action. However, we do not rely on our memory completely and we make notes and store important data and information using other media such as notebooks, manual, journal document, etc. Similarly, computers have two types of memory - primary and secondary.

1. Primary Memory: Primary memory is an essential component of a computer system. Program and data are loaded into the primary memory before processing. The CPU interacts directly with the primary memory to perform read and write operation. It is of two types.

1. Random Access Memory (RAM)

2. Read Only Memory. (ROM)

RAM: RAM is volatile. That is, as long as the power is supplied to the computer, it retains the data in it. But as soon as the power supply is turned off, all the contents of RAM are wiped out. It is used to store data temporarily while the computer is working. Whenever the computer is started or a software application is launched, the required program and data are loaded into RAM for processing. RAM is usually referred to as main memory and it is faster than the secondary memory or storage devices.

ROM: ROM is non-volatile, which means its contents are not lost even when the power is turned off. It is used as a small but faster permanent storage for the contents, which are rarely changed. For example, the start-up program (boot loader) that loads the operating system into primary memory is stored in ROM.

2. Cache Memory: RAM is faster than secondary storage, but not as fast as computer processor. So because of RAM, a CPU may have to slow down. To speed up the operations of the CPU, a very high speed memory is placed between the CPU and the primary memory known as cache. It stores the copies of the data from frequently accessed. Primary memory locations, thus reducing the average time required to access data from the primary memory. When the CPU needs some data, it first examines the cache. In case the requirement is met, it is read from the cache, otherwise the primary memory is accessed.

3. Secondary Memory: Primary memory has limited storage capacity and is either volatile (RAM) or read only memory (ROM). Does a computer system needs auxiliary or secondary memory to permanently store the data or instructions for a future use. The secondary memory is non-volatile and has a larger storage capacity than primary memory. It is slower and cheaper than the main memory, but it cannot be accessed directly by the CPU. Contents of secondary storage needs to be first brought into the main memory of the CPU to access. Examples of secondary memory devices include hard disk drive (HDD) CD / DVD memory card. Solid state device (SSD) etc.



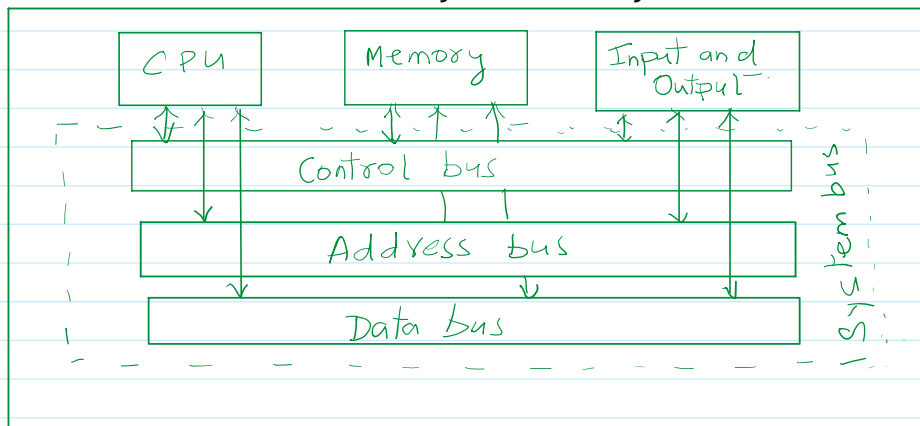
Data transfer between Memory and CPU

Data needs to be transferred between the CPU and primary memory, as well as between primary and secondary memory.

Data are transferred between different components of a computer system using physical wires called bus. For example, bus is used for data transfer between AUSB port and hard disk or between a hard disk and main memory. Bus is of three types.

- i) Data bus to transfer data between different components.
- ii) Address bus to transfer addresses between CPU and the main memory. The address of the memory location that the CPU wants to read or write from is specified in the address bus.
- iii) Control bus to communicate control signals between the different components of a computer.

All these three busses collectively make the system bus.



Data transfer between components through system bus.

As the CPU interacts directly with main memory, any data entered from the input device or the data to be accessed from hard disk or SSD needs to be placed in the main memory for further processor. The data is then transferred between CPU and main memory using bus.

The CPU places on the address bus the address of the main memory location from which it wants to read data or to write data. While executing the instructions, the CPU specifies the read or write control signal through the control bus.

As the CPU may require to read data from main memory or write data to main memory, a data bus is bidirectional. But the control bus and address bus are unidirectional. To write data into memory, the CPU places the data on the

data bus, which is then written to the specific address provided through the address bus. In case of read operation, the CPU specifies the address and the data is placed on the data bus by a dedicated hardware called memory controller. The memory controller manages the flow of data into and out of the computer's main memory.

Microprocessors

In early days, a computer CPU is used to occupy a large room or multiple cabinets. However with advance. In technology, the physical size of CPU has reduced and now possible to place a CPU on a single microchip only. A processor (CPU) which is implemented on a single microchip is called microprocessor. Nowadays, almost all Cpus are microprocessors. Hence the term are used synonymously for practical purposes.

Microprocessor is a small sized electronic component inside a computer that carries out various task involved in data processing as well as arithmetic and logical operations. These days a microprocessor is a built over an integrated circuit comprising millions of small components like registers, transistors and diodes.

Microprocessors have evolved overtime in terms of their increasing processing capability, decreasing physical size and reduced cost. Currently available microprocessors are capable of processing millions of instructions per second.

Here's a structured table showing the evolution of microprocessors across generations, including chip type, word size, maximum memory, clock speed, cores, and examples. This captures the major milestones from the 1970s to modern multi-core processors.

Microprocessor Generations Overview

Generation Era	Chip Type	Word Size	Max Memory Size	Clock Speed	Core s	Examples
First Gen (1971-1972)	SSI/MSI ICs	4-bit	~640 bytes	740 kHz	1	Intel 4004, Rockwell PPS-4
Second Gen (1973-1978)	NMOS	8-bit	64 KB	500 kHz – 3 MHz	1	Intel 8080, 8085, Motorola 6800
Third Gen (1979-1980)	HMOS	16-bit	1 MB	4-10 MHz	1	Intel 8086, 80286, Motorola 68000
Fourth Gen (1981-1995)	HMOS/CMOS	32-bit	4 GB	16-100 MHz	1	Intel 80386, 80486, Motorola 68020
Fifth Gen (1995-2005)	CMOS	64-bit	16 EB (theoretical)	66 MHz – 3 GHz	1-2	Intel Pentium, AMD Athlon
Modern Era (2006-Present)	CMOS, Multi-core	64-bit	>16 EB	1.2 GHz – 6 GHz	2-64 +	Intel Core i3/i5/i7/i9, AMD Ryzen, Apple M1/M2

Key Highlights

- **First Generation (Intel 4004, 1971):** 4-bit processor, only 640 bytes of memory, single-core at 740 kHz.
- **Second Generation (Intel 8085, 1976):** 8-bit, 64 KB memory, 3 MHz clock speed.
- **Third Generation (Intel 8086, 1978):** 16-bit, 1 MB memory, 10 MHz clock speed.

- **Fourth Generation (Intel 80386, 1986):** 32-bit, 4 GB memory, up to 33 MHz.
- **Fifth Generation (Pentium, 1993):** 64-bit, multi-core beginnings, GHz-level speeds.
- **Modern Era (Core i-series, Ryzen, Apple M1):** Multi-core (up to 64+), speeds beyond 5 GHz, massive memory addressing.

Risks & Trade-offs

- **Heat & Power Consumption:** Higher clock speeds and more cores increase thermal output, requiring advanced cooling.
- **Backward Compatibility:** Older software may not run efficiently on modern architectures.
- **Cost vs. Performance:** Cutting-edge chips (Apple M2 Ultra, Intel Core Ultra) are expensive compared to mid-range CPUs.

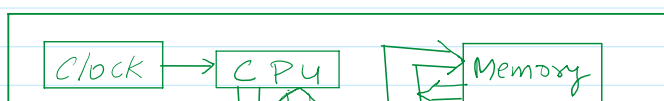
Microprocessor Specifications

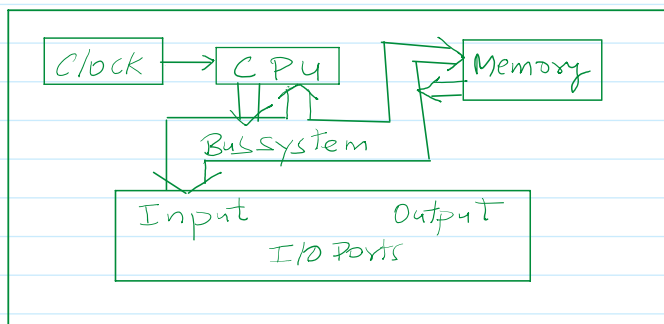
Microprocessors are classified on the basis of different features, include chip type, word size, memory size, clock speed, etcetera. These features are briefly explained below:

1. **Word size:** Word size is the maximum number of bits that a microprocessor can process at a time. Earlier a word was of eight bits as it was the maximum limit at that time. At present, the minimum word size is 16 bits and the maximum word size is 64 bits.
2. **Memory Size:** Depending upon the word size, the size of RAM varies. Initially RAM was very small (4MB,) Due to 4/8 bits word size. As word size increase to 64 bits, it has become feasible to use RAM of size up to 16 exabytes (EB)
3. **Clock speed:** Computers have internal clock that generates pulses (signals) at regular intervals of time. Clock speed simply means the number of pulses generated per second by the clock inside a computer. The clock speed indicates the speed at which the computer can execute instructions. Earlier it was measured in Hz(Hertz) and KiloHertz(kHz). But the advancement in technology and chip density? It is now measured in gigahertz (GHz), i.e., Billions of pulses per second.
4. **Cores:** Core is a basic computation unit of the CPU. Earlier processor had only one computation unit, thereby capable of performing only one task at a time. With the advent of multicore processors, it has become possible for the computer to execute multiple tasks, thereby increasing the systems performance. CPU with 2, 4 and eight cores are is called dual core, quad core and Octa core processors respectively.

Microcontrollers

The microcontroller is a small computing device which has a CPU of fixed amount of RAM, ROM and other peripherals all embedded on a single chip as compared to microprocessor that has only a CPU on the chip. The structure of a microcontroller is shown in the following figure:





Structure of Microcontroller

Keyboard, mouse, washing machine, digital camera, pen drive, remote controller, microwave are few examples of microcontrollers. As these are designed for a specific task only, hence their sizes as well as cost is reduced.

Because of the very small size of the microcontroller, it is embedded in another device or system to perform a specific functionality. For example, the microcontroller in a fully automatic washing machine is used to control the washing cycle without any human intervention. The cycle starts with the filling of water, after which the clothes are soaked and washed. Thereafter the water is drained and the clothes are spin dry. The simple use of microcontroller has permitted repetitive execution of tedious task automatically without any human intervention, thereby saving precious time.

DATA AND INFORMATION

A computer is primarily for processing data. A computer system considers everything as data, be it instructions, pictures, songs, videos, documents, etc. Data can also be raw and unorganized facts that are processed to get meaningful information.

So understanding the concept of data along with its different types is crucial to understand the overall functioning of a computer. Sometimes people use the terms data, information and knowledge interchangeably, which is incorrect.

Data and its Types

A computer system has many input devices which provide it with raw data in the form of facts. Concepts, instructions, etc., Internally everything is stored in binary form (0 and 1) But externally data can be input to a computer in the text form consisting of English alphabets, A-Z, a-z, numerals 0-9, and special symbols like @, #, etc. Data can be input in other languages too, or it can be read from the files. The input data may be from different sources, hence it may be in different format. For example, an image is a collection of red, green, blue (RGB) pixels. A video is made up of frames and a fee receipt is made of numeric and non-numeric characters. Primarily, there are three types of data.

1. **Structured Data:** Data which follows a strict record structure and is easy to comprehend is called structure data. Search data with PRE - specified. Tabular format may be stored in a data file to access in the future. The following table shows a structure data related to monthly attendance of

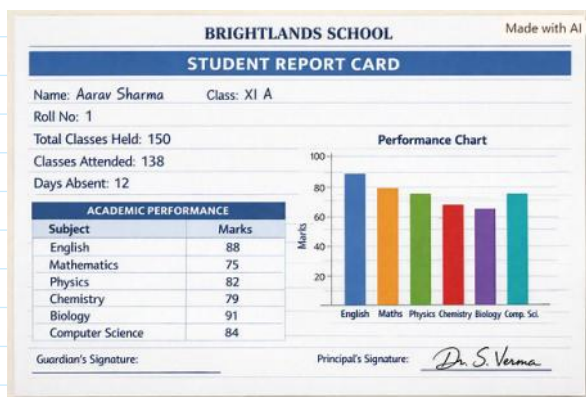
students maintained by the student.

Roll No	Name	Month	Attendance %
101	Ananya Sharma	July	92%
102	Rahul Mehta	July	85%
103	Priya Das	July	96%
104	Arjun Singh	July	78%
105	Sneha Roy	July	88%

It is clear that such data is organized in row / column format and is easily understandable. Structured data may be sorted in ascending or descending order. In the example, attendance data is sorted in increasing order on the column "Roll No". Other example of structured data includes Sales transaction online, railway ticket booking, ATM transactions, etc.

Unstructured Data

Data which are not organized in a predefined record format is called unstructured data. Examples include audio and video files, graphics, text documents, social media posts, satellite images, etc.



Unstructured Data

Semi-structured Data

Data which have no. Defined structure but maintains internal tags or marking two separate data elements are called semi structured data. Examples include email document, HTML page, separated values (CSV file) etc.

CSVDataFiles > XYZHotels.csv	
1	SunShine Beach Resort,SunVille Beach Resort,Sunny Beach Resort
2	4.75,3,4.5
3	2.5,4,2
4	3.5,2.5,3
5	4,2,3.5
6	1.5,4.5,1

Ratings of different resort are shown above.

Data Capturing, Storage and Retrieval

To process data we need to first input or capture the data. This is followed by storage in a file or a database so that it can be used in the future. Whenever data is to be processed, it is first retrieved from the file or database so that we can perform further actions on it.

- 1. Data Capturing:** It involves the process of gathering data from different sources in the digital form. Data capturing may vary from instruments like keyboard barcode readers used at shopping outlets. Comments or posts over social media, remote sensors on the earth, orbiting satellites, etc. Sometimes heterogeneity among data sources makes data capturing a complex task.
- 2. Data Storage:** It is the process of storing the captured data for processing later. Nowadays, data is being produced at a very high rate and therefore data storage has become a challenging task. However, the decrease in the cost of digital storage devices has helped in simplifying this task. There are numerous digital storage devices available in the market. Data keeps on increasing with time hence that storage devices also required to be upgraded periodically. In large organizations, computers with large and faster storage are called data servers. Are deployed to store vast amount of data. Such dedicated computers helps in processing data efficiently. However, the cost (both hardware and software) Of setting up a data server as well as its maintenance is high, especially for small organization and startups.
- 3. Data Retrieval:** It involves fetching data from storage devices for its processing as per the user requirement. As databases grow, the challenges involved in search and retrieval of the data in acceptable time also increase. Minimizing data access time is crucial for faster data processing.

Data Deletion and Recovery

One of the biggest threats associated with the digital data is its deletion. The storage devices can malfunction or crash down in the deletion of data stored. Users can accidentally erase data from storage devices, or a hacker or malware can delete the digital data intentionally.

Deleting digitally stored data means changing the details of data a bit level which can be very time consuming. Therefore, when any data is simply deleted, its address entry is marked as free and that much space is shown as empty to the user without actually deleting the data.

In case data gets deleted accidentally or corrupted then arises a need to recover the data. Recovery of data is possible only if the contents or memory space marked as deleted have not been overwritten by some other data. Data recovery is a process of retrieving deleted, corrupted, and lost data from secondary storage devices.

There are usually two security concern associated with data. One is its deletion by some unauthorized person or software. These concerns can be avoided by limiting access to the computer system and using passwords for user accounts and files wherever possible. There is also an option of encrypting files to protect them from unwanted modification.

The other concern is related to unwanted recovery of data by unauthorized user or software. Many a times we discard our old, broken or malfunction storage devices without taking care to deleted data. We assume that the contents of deleted files are permanently removed. However, if these storage devices fall into the hands of mischief - mongers, they can easily recover data from such devices. This poses a threat to data confidentiality. This concern can be mitigated by using proper tools to delete or share data before disposing of any old or faulty storage device.

SOFTWARE

Till now we have studied about the physical components or the hardware of the computer system. But the hardware is of no use on its own. Hardware needs to be operated by a set of instructions. These sets of instructions are referred to as software. It is that component of a computer system which we cannot touch or view physically. It comprises of the instructions and data to be processed using the computer hardware. The computer software and hardware complete any task together.

The software comprise a set of instructions which on execution deliver the desired outcome. In other words, each software is written for some computational purpose. Some example of software include operating systems like Ubuntu, Mac, OS, X, Windows 11. Word processing tools like Libreoffice or Microsoft Office. Video players like VLC Player, photo editors like Jim and Libre Office Draw. A document or image stored on the hard disk or pen drive is referred to as a soft copy. Once printed, the document or an image is called a hard copy.

Need of Software

The sole purpose of a software is to make the computer hardware useful and operational. A software knows how to make different hardware components of a computer work and communicate with each other as well as with the end user. We cannot instruct the hardware of a computer directly. Software act as an interface between human user and the hardware.

Depending on the mode of interactions with hardware and functions to be performed, the software can be broadly classified into three categories:

- i) **System Software**
- ii) **Programming Tools**
- iii) **Application Software**

System Software: *The software that provides the basic functionality to operate a computer by interacting directly with its constituent hardware is termed as system software.*

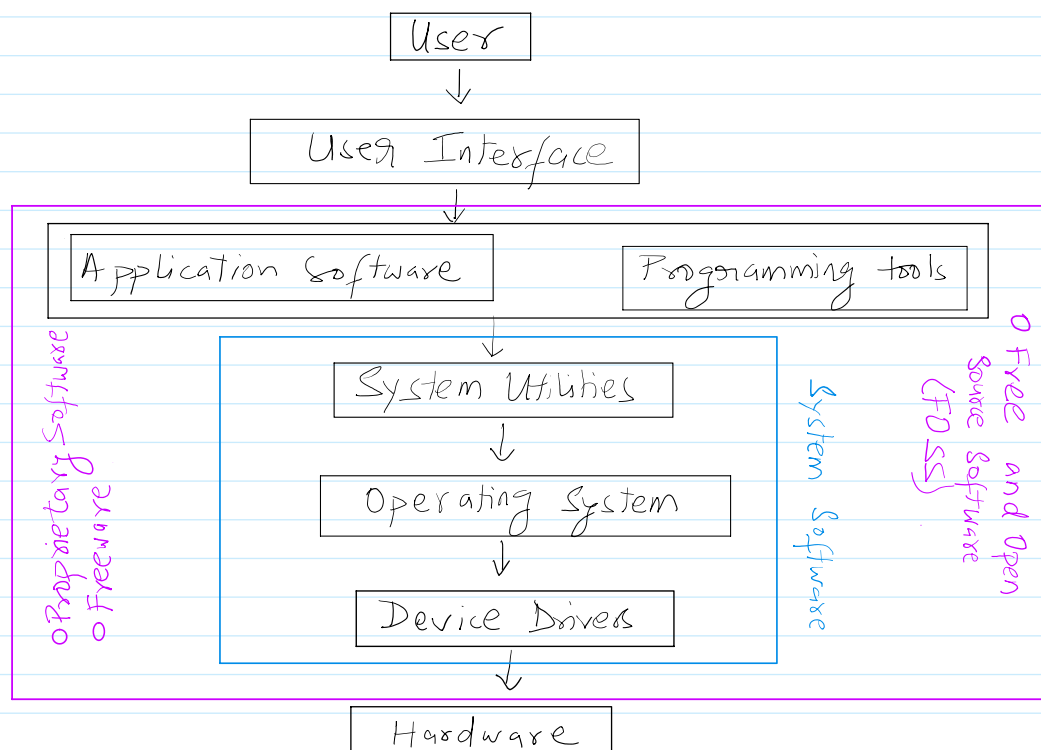
A system software knows how to operate and use different hardware components of a computer. It provides services directly to the end user or to some other software. Examples of system software include operating system, system utilities, device drivers, etc.

A. Operating System: As the name implies, the operating system is a system software that operates the computer. An operating system is the most basic system software, without which other software cannot work. The operating system manages other application programs and provide access and security to the users of the system. Some of the popular operating systems are Windows, Linux, Macintosh, Ubuntu, Fedora, Android, iOS, etc.

B. System Utilities: Software used for maintenance and configuration of the computer system is called System utility. Some system utilities are shipped with the operating system, for example disk defragmentation tool. Formatting utility, system restore utility, etc. Another set of utilities are those which are not shipped with the operating system but are required to improve the performance of the system. For example, antivirus software, disk cleaner tool, disk compression software, etc.

C. Device Drivers: As the name signifies, the purpose of a device driver is to ensure proper functioning of a particular device. When it comes to the overall working of a computer system, the operating system does the work. But everyday new devices and components are being added to a computer system. It is not possible for the operating system alone to operate all of the existing and new devices where each device has diverse characteristics. The responsibility for overall control, operation and management of a particular device at hardware level is delegated to its device driver.

The device driver acts as an interface between the device and the operating system. It provides required services by hiding the details of operations performed at the hardware level of the device. Just like a language translator, a device driver acts as a mediator between the operating system and the attached device.



Categorisation of software

Programming Tools

In order to get some work done by the computer, we need to give instructions which are applied on the input data to get the desired outcome. Computer languages are developed for writing these instructions. It is important to understand here that computers and human understand completely different languages. While humans are able to write programs in high level language, computers understand machine language. There is a continuous need for

conversion from high level to machine level language for which translators are needed. Also to write the instructions code editors example (VS Code is used as an editor for many programming languages).

Classification of programming languages

It is very difficult for a human being to write instruction in the form of ones and zeros. So different types of computer programming languages are developed to simplify the coding. Two major categories of computer programming languages are low level languages and high level languages.

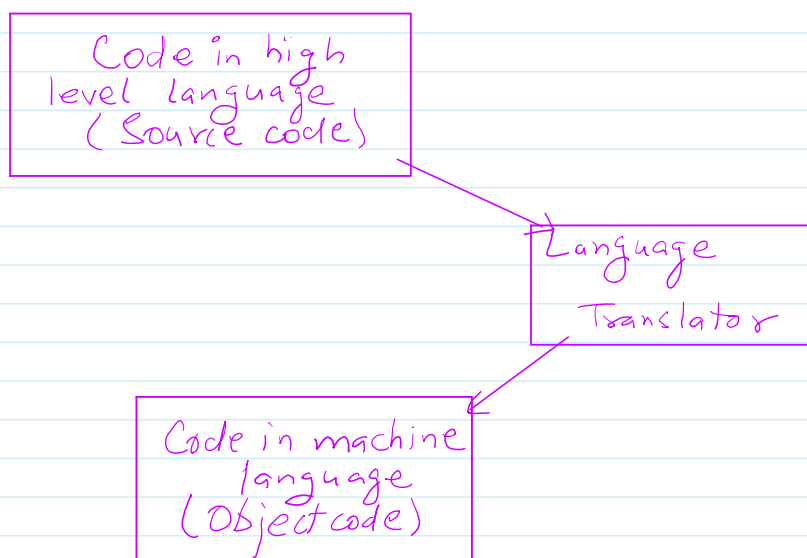
Low level languages are machine dependent languages and include machine language and assembly language. Machine language uses ones and zeros to write instructions which are directly understood and executed by the computer. But writing a code in machine language is difficult as one has to remember all operation codes and machine addresses. Also, finding errors in the code written in machine language is difficult.

To simplify the writing of code, assembly language was developed that allowed uses of English - like words and symbols instead of ones and zeros. But one major drawback of writing code in this language is that the code is computer specific. That is, the code written in one type of CPU cannot be used for another type of CPU.

High level languages are machine independent and are simpler to write code into. Instructions are using English like sentences and each high level language follows a set of rules similar to natural languages. However, these languages are not directly understood by the computers. Hence translator are needed to translate high level language codes into machine language. Examples of high level language include C++, Java, Python, etc.

Language Translators

As the computer can understand only machine language, a translator is needed to convert program written in assembly or high level language to machine language. The program code written in assembly or high level language is called source code. The source code is converted by a translator into a machine understandable form called object (machine) code.



(Object code)

Translator to convert source code into object code.

As we have different types of computer languages, different translators are needed to convert the source code into machine code. The three types of translator used in computing systems are assembler, compiler and interpreter.

The translator used to convert the code written in assembly language to machine language is called assembler. Each assembler can understand a specific microprocessor instruction set only and hence the machine code is not portable.

We also need translator to convert code written in high level language (source code) to machine understandable form (machine code) for execution by the computer. **Compilers** convert the source code into machine code. If the code follows all syntactic rules of the language, then it is executed by the computer. Once translated, the compiler is not needed.

An **interpreter** translates one line at a time instead of the whole program at one go. Interpreter takes one line, convert it into executable code if the line is syntactically correct, and then it repeats these steps for all lines in the source code. Hence interpreter is always needed when a source code is to be executed.

Program Development Tools

Whenever we decide to write a program, we need a text editor. An editor is a software that allows us to create a text file where we typed instructions and store the file as the source code. Translator is used to get the object code for execution. In order to simplify the program development, there is a software called Integrated Development Environment. (Called IDE) Consisting of text editor, building tools and debugger. A program can be typed, compiled and debugged from the IDE directly. Beside Python IDLE. Netbeans, Eclipse, Atom. VS code are few other examples of IDE 's. Debugger, as the name implies, is the software to detect and correct errors in the source code.

Application Software

The system software provides the core functionality of the computer system. However, different users need the computer system for different purposes depending upon their requirements. Hence, a new category of software is needed to cater to different requirements of the end - users. This specific software works on the top of the system. Software is termed as application software. There are again two broad categories of application software - a general purpose and customized application software.

General purpose software

The application software developed for generic applications to cater a big audience in general are called general purpose software. Such ready made application software can be used by end users as per their requirements. For

example, spreadsheet tool Calc of Libreoffice can be used by any computer user to do calculations or to create account sheet. Adobe Photoshop Gym. Mozilla, web browser, iTunes etc. fall in the category of general purpose software.

Customized Software

These are custom or tailor made application software that are developed to meet the requirements of a specific organization or an individual. They are better. Routed to the needs of an individual or an organization, considering that they are designed as per special requirements. Some examples of user defined software include websites, school management software, accounting software, etc. It is similar to buying a piece of cloth and getting a tailor made garment with the fitting color and fabric of our choice.

Proprietary or free and open source software

The developers of some application software provide their source code as well as software freely to the public with the aim to develop and improve further with each other's help. Such software is known as free and open source software (FOSS) For example, the source code of the operating system Ubuntu is freely accessible for anyone with the required knowledge to improve or add a new functionality. More examples of force includes Python, Libreoffice, Openoffice, Mozilla Firefox, etc. Sometimes software are freely available for use. But source code may not be available. Such software's are called freeware. Examples of freeware are Teams, zoom, Adobe PDF reader, etc. When the software is to be used, has to be purchased from the vendor who has the copyright of the software, then it is a proprietary software. Examples of proprietary software include Microsoft Windows, Tally, Quickheal, etc. A software can be freeware or open source or proprietary software, depending upon the terms and conditions of a person or group who has developed and released that software.

Operating System

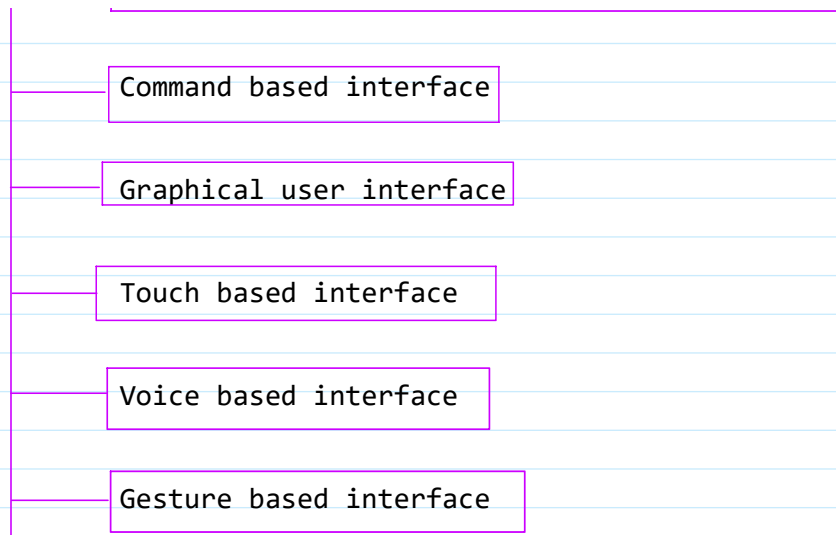
An operating system (OS) can be considered to be a resource manager which manages all the resources of a computer that is its hardware including CPU, RAM, disk, network and other input - output devices. It also controls various application software and device drivers manager, just system security and handles access by different users. It is the most important system software. Examples of popular OS are Windows, Linux, Android, Macintosh and so on.

The primary objectives of an operating system are two - fold. The first is to provide the services for building and running application programs. When an application program needs to be run, it is the operating system which loads that program into memory and allocates it to the CPU for execution. When multiple application program needs to run, the operating system decides the order of the execution.

The second objective of an operating system is to provide. An interface to the user through which the user can interact with the computer. A user interface is a software component which is a part of the operating system and whose job is to take commands or input from the user for the operating system to process.

Types of user interface of operating system

Command based interface



Types of user interface of OS.

OS User Interface

There are different types of user interface, each of which provides different functionality.

- 1. Command - based interface:** A command - based interface requires a user to enter the commands to perform different tasks like creating, opening, editing or deleting a file etc. The user has to remember the names of all such programs or specific commands which the operating system supports. The primary input device used by the user for the command based interface is the keyboard. Command based interface is often less interactive and usually allows a user to run a single program at A time. Examples of operating system with command based interface include MS-DOS and UNIX.
- 2. Graphical User Interface:** Graphical user interface (GUI) lets users run programs or give instructions to the computer in the form of icons, menus and other visual options. Icons usually represents files and programs stored on the computer, and Windows represents running programs that the user has launched through the operating system. The input devices used to interact with the GUI commonly include the mouse and the keyboard. Examples of operating system with GUI interfaces include Microsoft Windows, Ubuntu, Fedora and Macintosh, among others.
- 3. Touch - based interface:** Today's smart phones, tablets and PCs allow users to interact with the system simply using the touch input. Using the touch screen, a user provides input to the operating system which are interpreted by the OS as commands like opening an app, closing an app, dialing a number, scrolling across app, etc. Examples of popular operating system with touch based interfaces are Android and iOS. Windows 11 also supports touch based interfaces on touch screen devices.
- 4. Voice-based Interface:** Modern computer designed to address the needs of all types of user, people with special needs and people who want to interact with computer or smartphones while doing some other task. For

users who cannot use the input devices like the mouse, keyboard and touch screen, modern operating system provide other means of human - computer interaction. Users today can use voice based command to make a computer work in a desired way. Some operating system which provide voice based control to users include iOS (Siri) Android (Google Now or OK Google.) Microsoft Windows 11 (Cortana) and so on.

- 5. Gesture-based Interfaces:** Some smartphone based on Android and IOS as well as laptops let's user interact with devices using gestures like waving, tilting, eye motion and shaking. This technology is evolving faster and it has promising potential for applications in gaming, medicine and other areas.

Functions of Operating System

- 1. Process Management:** When a computer system is operational, different tasks are running simultaneously. A program is intended to carry out various task. A task in execution is known as process. We can activate a system monitor program that provides information about the processes being executed on a computer. In some systems it can be activated using Ctrl+Alt+Delete . It is the responsibility of the operating system to manage these processes and get multiple tasks completed in minimum time. As CPU is the main resource of the computer system, its allocation among processes is the most important service of the operating system. Hence, process management concerns the management of multiple processors, allocations of required resources and exchange of information among processes.
- 2. Memory Management:** Primary or main memory of a computer system is usually limited. The main task of memory management is to give (allocate) and take (free) memory from the running processes. Since there are multiple processes running at a time, their arises are need to dynamically (on - The - go) Allocate and free memory to the processes. Operating system should do it without affecting other processes that are already residing in the memory. And once the process is finished, it is again the responsibility of the operating system to take the memory space back for the reutilization. Hence memory management concerns with the management of main memory. So the maximum memory is occupied or utilized by a large number of processes. While keeping the track of each and every location within the memory as free or occupied.
- 3. File Management:** Data and programs are stored as files in the secondary storage of a computer system. File management involves the creation of, deletion, deletion and protection of files in a secondary memory. Protection is a crucial function of an operating system, as multiple users can access and use a computer system. There must be a mechanism in place that will stop users from accessing files that belongs to some other users and have not been shared with them. File management system manager secondary memory while memory management system handles the main memory of the computer system.
- 4. Device Management:** A computer system has many I / O devices and hardware connected to it. Operating system manages these heterogeneous devices that are interdependent. The operating system interacts with the device driver and the related software for a particular device. The operating system

must also provide the options for configuring a particular device so that it may be used by an end user or some other device. Just like files, devices also need security measures and their access to different devices must be restricted to the operating system, to the authorized users, software and other hardware only.

